Design and Implementation of a Buck Converter for DC Voltage Regulation

Udara Deshan Tharusha Nimesh Dinusha Dissanayaka
Senuda Ratnayake

Department of Electronic & Telecommunication Engineering,
University of Moratuwa, Sri Lanka.

April 28, 2026

Introduction

Introduction

A buck converter is a type of DC–DC converter used to step down a higher DC voltage to a lower level efficiently.

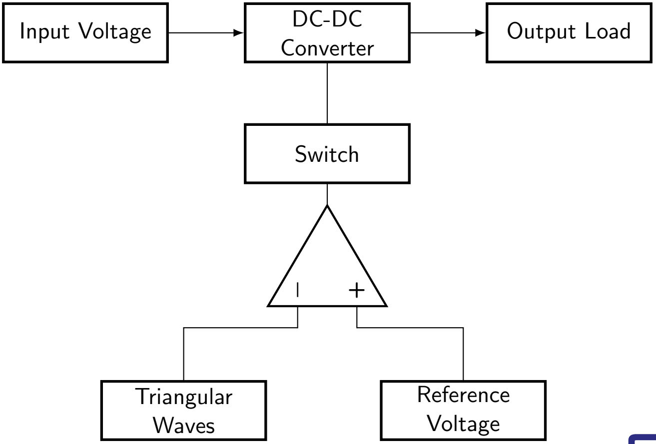
- Widely used for voltage regulation in electronic devices
- Uses switching techniques for high efficiency
- PWM generated using op-amp comparator and triangle wave
- Output voltage controlled by duty cycle

Device Specifications

Specifications

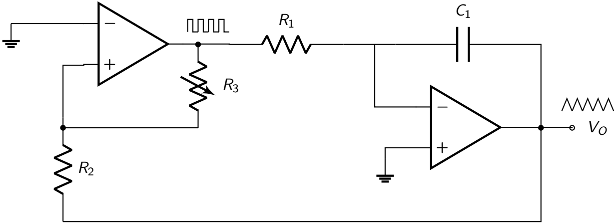
- Input Voltage: 15 V
- Output Voltage: 3V – 12 V
- Switching Frequency: 16 kHz
- Maximum Load Current: 1 A
- Duty Cycle Range: $D = 0.2$ to 0.8
- Target Inductor Ripple Voltage : 1 % of V_{out}
- Target Inductor Ripple Current : 10 % of I_{out}

Block Diagram



Circuit Implementation

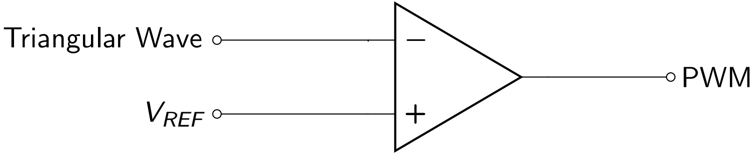
Triangular Waveform Generator



Operation

- Generated using an op-amp integrator and comparator
- Produces a continuous triangular waveform
- Used as carrier signal for PWM generation
- Compared with reference voltage to create PWM output

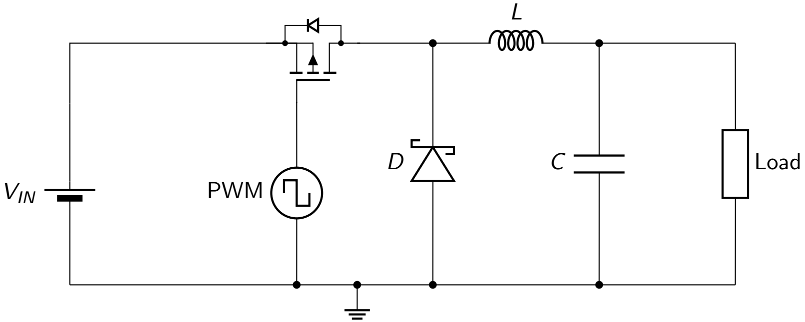
Comparator



Operation

- The comparator acts as a decision-maker by constantly comparing a triangular signal with a target control voltage.
- It outputs a high signal when the control voltage is greater than the reference and a low signal when it is less, converting the intersection points into PWM signal.

Buck Converter



Buck Converter Circuit

Circuit Operation

- Input DC voltage V_{in} is applied to MOSFET switch
- PWM signal controls ON/OFF operation of the switch
- Inductor stores energy during ON state
- Diode provides current path during OFF state
- LC filter smooths output and reduces ripple
- Output voltage depends on PWM duty cycle

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Component Selection & Justification

Op-Amp Selection

1. Slew Rate Requirement To generate square wave in PWM generator circuit,

Minimum Required Slew Rate

$$SR \geq |V_p \cdot j\omega e^{j\omega t}| = 7.5 \times 2\pi \times 16 \times 10^3 \times 10 = 7.5\ \mathrm{V\,\mu s^{-1}}$$

2. Gain Bandwidth Requirement To accurately reproduce a square wave, the op-amp must support higher frequency harmonics.

Minimum Required GBW

$$GBW \geq n \cdot f_0 = 10 \times 15\,\mathrm{kHz} = 150\,\mathrm{kHz}$$

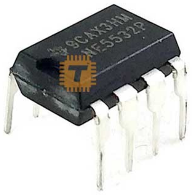
Note: Higher-order harmonics are required to approximate a square wave, which increases the effective slew rate requirement. In this design, up to the 10th harmonic is considered.

Op-Amp Selection (Continued)

3. Input Voltage Requirement $V_{CC}^{+} - V_{CC}^{-} \leq 15\ \mathrm{V}$

Selected Op-Amp : NE5532

- High slew rate (Typical $9\,\mathrm{V\,\mu s^{-1}}$).
- 6 to 40 supply voltage range.
- High gain bandwidth product (10 MHz).
- Low cost and readily available.



Switching Transistor Selection

1. Voltage Rating Higher than the input voltage, typically 1.2 to 1.5 (22.5 V).
2. Current Rating Should be able to handle peak current, so at least 1 A (2 A is recommended for safety).
3. Switching Voltage: 0 to -15 V

Selected MOSFET: IRF9540

- **Voltage Rating:** $V_{DS} = 100\,\text{V}$ ($>$ 22.5 V)
- **Current Rating:** $I_D = 19\,\text{A}$ ($>>$ 2 A)
- **Gate-source threshold voltage** -2 to -4 V.

Diode Selection

1. Voltage Rating: Higher than the input voltage, typically 1.2 to $1.5 \times V_{in}$ (22.5 V).
2. Current Rating: Should handle load current, at least 1 A.
3. Fast Switching: Required to reduce switching losses in PWM operation.

Selected Diode: 1N5819 (Schottky)

- **Reverse Voltage Rating:** $V_{RRM} = 40\,\mathrm{V}$ $(> \ 22.5\,\mathrm{V})$
- **Forward Current Rating:** $I_F = 1\,\mathrm{A}$
- **Forward Voltage Drop:** $V_F \approx 0.3\,\mathrm{V}$ to $0.45\,\mathrm{V}$ (Lower than standard $0.7\,\mathrm{V}$ for higher efficiency)

Inductor and Capacitor Selection

Inductor Selection

1. **Inductance:** Based on ripple current requirement
2. **Current Rating:** $I_L \geq I_{peak}$

Capacitor Selection

1. **Capacitance:** Based on output voltage ripple
2. **Low ESR:** Reduces ripple

Selected Inductor and Capacitor

- $L = 2.2\,\text{mH}$
- $C = 33\,\mu\text{F}$

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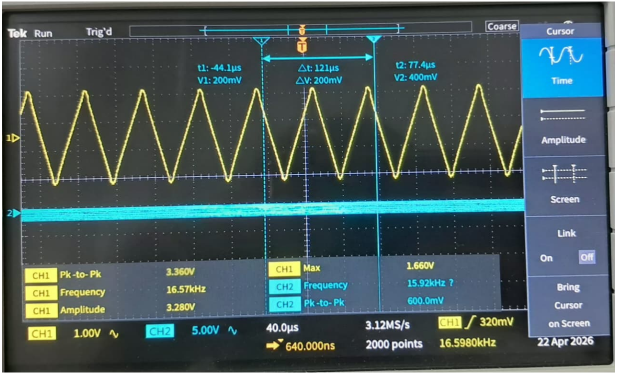
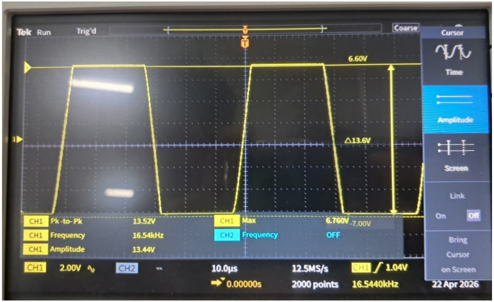
Results & Comparison

Triangular Waveform Generation



- The obtained frequency was 16.57 kHz. (< 20 kHz)
- Actual $V_{pp} = 3.36$ V. (< 4 V)

PWM Signal Generation



- $V_{pp} = 13.6$ V and $f = 16.54$ kHz.
- Practical Op-Amp slew rate $= \frac{13.52\,\text{V}}{5.8\,\mu\text{s}} = 2.33\,\text{V}\,\mu\text{s}^{-1}$. This is much slower than the expected value.
- 16.79% Lost in Duty cycle.

Thank you!